INDIVIDUAL WEEKLY PROJECT REPORTS

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Role: Exploratory Data Analysis and Data Visualization
Project Title: Flight Delay Analysis and Visualization

WEEK 3 – Understanding Data Cleaning and Preparing for EDA

During the third week, I focused on understanding how the data cleaning process affected the quality of the visualizations. Since graphs and charts are generated from the available data, errors in the dataset could result in misleading visual representations.

I coordinated with Annalakshmi to understand the preprocessing activities being carried out. I learned about the importance of handling missing values, checking duplicates, and identifying invalid numerical entries before creating visualizations.

I also explored how outliers could influence graphs such as histograms and box plots. For example, unusually large delay values could affect the scale of a graph and make the distribution of ordinary observations harder to interpret.

Another part of my work involved identifying the variables that could be used after preprocessing. I considered the types of comparisons that could be performed using airline information, departure delay, arrival delay, and airport-related attributes.

I also began organizing the possible visualizations into categories, such as distribution analysis, comparison analysis, and relationship analysis.

Outcome and Learning:

This week helped me understand the connection between data preprocessing and visualization. I learned that the accuracy and clarity of a chart depend heavily on the quality of the underlying data.